

Zach Taylor

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SUMMARY

Software engineer with strong interests in **low-level systems**, **AI**, and application development in **C++** and **Python**. Eager to learn, grow, and expand my expertise while contributing to innovative projects that give value to customers.

SKILLS

C++, Python, Typescript, Next.js, TensorFlow, CUDA, GitHub, Linux, Jenkins, AWS, Docker, FPGA

EDUCATION

Bachelor of Science - Oregon State University

September 2019 – June 2023

Winner of the Pioneering Concept Award at the 2023 OSU Engineering Expo

EXPERIENCE

Software Engineer Intern *Rohde & Schwarz*

April 2024 – Jan 2025

- Created tailored network analysis solutions using **C++** for customers, enabling them to garner more valuable diagnostics on their business infrastructure.
- Designed a composite **API** to replace slow, inefficient code and improve long-term maintainability into new and existing software products.
- Identified and resolved **15** critical computation errors affecting all software applications through **unit testing**, maintaining software quality and saving thousands in potential damages.
- Automated **SSH** together with Ethernet configuration using **Python**, shortening test and development time by **8%** over the next several sprints.

Undergraduate Teaching Assistant *Oregon State University*

September 2020 – June 2023

- Delivered personalized one-on-one and small group tutoring focused on reinforcing core concepts and addressing individual challenges, raising academic performance by **30%** among struggling students.
- Taught fundamental software courses including data structure and operating systems using **C** as well as full stack web development with **JavaScript**.
- Proctored exams containing **100-150** students, upholding academic integrity and ensuring a fair, distraction free testing environment.

Software Development Intern *Portland State University*

June 2019 – September 2019

- Streamlined food processing for the college of agriculture by designing a **TensorFlow** neural network model using **Python** with **OpenCV** for use on assembly lines.
- Optimized the model using a transfer learning approach, resulting in a **97%** detection accuracy through measured tests on factory assembly lines.
- Coordinated weekly meetings with engineers as well as end users to align product direction with objectives and keep development on schedule.

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, AI*

[GitHub](#)

- Developed an **AI** tracking robot to reduce falls among elderly patients in hospitals and nursing homes, utilizing **Python**, **OpenCV**, and **MediaPipe** libraries.
- Increased the fall detection accuracy by **56%** through live model training in local nursing homes along with creating custom MediaPip functions for figure tracking.
- Directed development efforts in a team of **7-8** engineers by mentoring members on AI deployment and performance optimization.
- Displayed progress to Intel sponsors during weekly meetings and presented the project at the 2023 OSU engineering expo.

SIMD Image Filter | *C++, SIMD*

[GitHub](#)

- Implemented a **C++** image filter library to learn the fundamentals of **SIMD** uses cases and architecture.
- Features blur, sharpen, and edge detection algorithms with metrics to show a **5x-7x** difference between SIMD and scalar applications.